

PatchSentinel

Autonomous DevSecOps & Adversary Remediation Agent

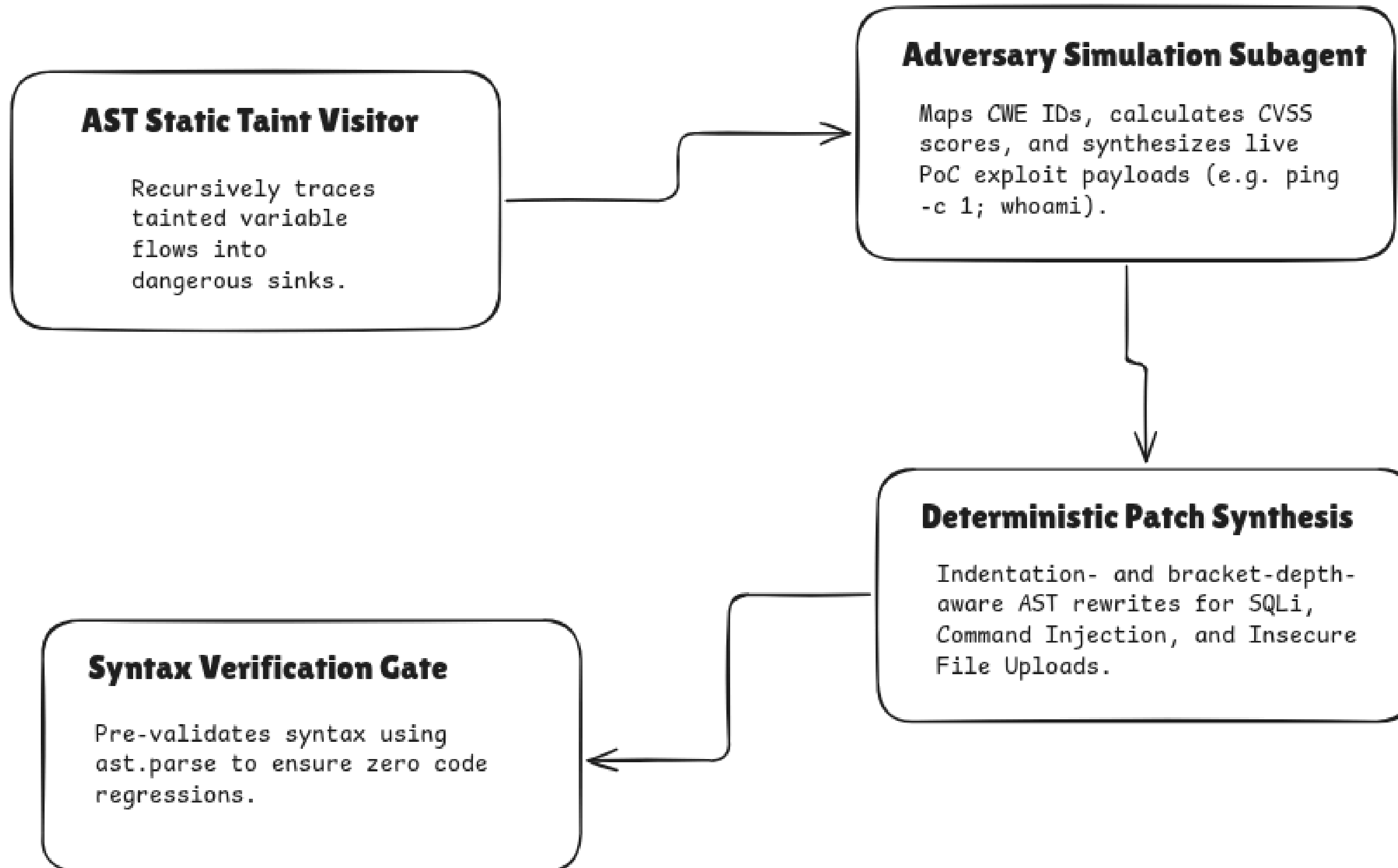
Real-World Adversary Simulation & Deterministic Code Patching

- Built with **IBM Bob 2.0** in **Agent** Mode
- 147 Unit Tests Passing
- Shift-Left Security Gate

The SAST Bottleneck

- **Alert Fatigue:** Over 70% of static warnings are **noisy**, **unexploitable**, or **false positives**.
- **Zero Exploit Evidence:** Abstract **CWE** tags give developers no proof of actual vulnerability.
- **Remediation Drag:** Industry MTTR exceeds 4 hours per flaw due to manual patch boilerplate.
- **Fragile AI Fixes:** Generic LLM completions frequently **break** syntax and indentation.

Dual-Engine Architecture



Shift-Left & Enterprise CI/CD

- **Git Pre-Commit Gate:** `python3 main.py --install-hook` halts insecure commits on localhost.
- **GitHub Actions SARIF Telemetry:** Direct **SARIF v2.1.0** ingestion into GitHub CodeQL and PR checks.
- **High-Assurance Reliability:** **147 unit tests** executing in **0.35 seconds** with **zero** regressions.

Engineered with IBM Bob 2.0

- **Full Agent Mode Lifecycle:** Scaffolded **AST** visitors, patch calculators, and CLI reporters.
- **8 Documented Milestones:** Full proof of task completions and Bobcoins tracked in **/bob_sessions**.
- **Autonomous Problem Solving:** Diagnosed **YAML** CI bugs, test-directory filters, and environment ignores.
- **Bottom Line:** Reduces vulnerability remediation time from 4 hours to under 10 seconds.

Thank You

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